Volume of pyramids and cones

1

a $160\ \text{cm}^3$ **b** $200\ \text{cm}^3$ **c** $154.78\ \text{cm}^3$ **d** $100\ \text{cm}^3$

e $282.33\ \text{mm}^3$

2 $\frac{448}{3}\ \text{cm}^3$

3 $2\,561\,328\ \text{m}^3$

4 $12\ \text{cm}$

5

a

i $128.45\ \text{m}$ **ii** $85\,633.33\ \text{m}^3$

b

i $13.92\ \text{cm}$ **ii** $125.3\ \text{cm}^3$

6 $2\,574\,466.7\ \text{m}^3$

7

a $4608\ \text{mm}^3$ **b** $3969\ \text{mm}^3$

8 $26\ \text{mm}$

9 $18\ \text{cm}$

10

a $25.13\ \text{cm}^3$ **b** $32.45\ \text{cm}^3$ **c** $56.55\ \text{units}^3$

11

a $220.0\ \text{cm}^3$ **b** $452.4\ \text{cm}^3$ **c** $1187.5\ \text{cm}^3$

12

a $43.37\ \text{m}^3$ **b** $36.65\ \text{cm}^3$

13

a $871.27\ \text{cm}^3$ **b** $103\,823.35\ \text{m}^3$

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14

a $16.0\ \text{m}$ **b** $25\ \text{cm}$

15

a $19.9\ \text{cm}$ **b** $8\ \text{mm}$ **c** $5.72\ \text{mm}$

16 $h = \sqrt[3]{H^3 \frac{x}{V}}$

Additional questions

17

a $440\ \text{cm}^3$ **b** $24.44\ \text{in}^3$ **c** $84.82\ \text{cm}^3$ **d** $1526.81\ \text{km}^3$

18 $55\ \text{in}^3$

19 6 units

20 $45\ \text{m}^2$

21 Yes, the silo has a height of $10\ \text{ft}$ which is shorter than the $12\ \text{ft}$ clearance of the doorway.

22 You can order either 3 of Option 1 or 1 of Option 2 to maximize the amount of kettlecorn you buy. Since the cone and the cylinder have the same diameter and height, three of the cones is equal to the volume of one cylinder. Both orders will cost $\$10.50$ total.

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